Impulse Control

Vedant Girnare

1 General Formulation

In this section, we consider an impulse control problem in which the underlying uncontrolled state process is a time-homogeneous diffusion, and the cost is of the infinite-horizon discounted form. Such a problem introduces the primary issues related to impulse control. Similar considerations apply in various other settings (e.g., time-dependent parameters, finite or indefinite time horizons).

We assume that, in the absence of control, the state process evolves as an uncontrolled diffusion:

$$dX(t) = b(X(t))dt + \sigma(X(t))dW(t), \quad X(0^-) = x \in \mathbb{R}^n.$$

We write $X(0^-)$ to emphasize that x is the state of the system at time 0 before any control is applied. An impulse control consists of a sequence of intervention times $0 \le \tau_1 < \tau_2 < \dots$ and impulse values $\zeta_1, \zeta_2, \dots$ taking values in some set Z. Given such a sequence, the controlled state process evolves as follows. For $0 \le t < \tau_1$, no control has been applied, yielding:

$$X(t) = x + \int_0^t b(X(s))ds + \int_0^t \sigma(X(s))dW(s), \quad 0 \le t < \tau_1.$$

Let $X(\tau_1^-)$ denote the state of the system immediately before time τ_1. At time τ_1, we apply an impulse ζ_1 to instantaneously move the state of the system from $X(\tau_1^-)$ to $X(\tau_1)$, where $X(\tau_1) = \Gamma(X(\tau_1^-), \zeta_1)$ and $\Gamma : \mathbb{R}^n \times Z \mapsto \mathbb{R}^n$ is the control function (a typical choice is $\Gamma(x, z) = x - z$). After time τ_1, no control is applied until time τ_2, so for $\tau_1 < t < \tau_2$ we have:

$$X(t) = X(\tau_1) + \int_{\tau_1}^t b(X(s))ds + \int_{\tau_1}^t \sigma(X(s))dW(s).$$

At time τ_2, we apply an impulse ζ_2 to instantaneously move the state from $X(\tau_2^-)$ to $X(\tau_2) = \Gamma(X(\tau_2^-), \zeta_2)$. The construction continues similarly for larger values of t.

Before placing the impulse control problem on rigorous footing, we note critical differences between an impulse-controlled state process and a classically controlled diffusion process, e.g., Equation (2.6.1) from Ross' notes. In classical control, we effectively modulate the values of b and σ, but we cannot directly adjust the actual values of the process. The opposite is true in impulse control: b and σ are fixed, but by choosing appropriate impulses, we directly influence the state. Furthermore, continuous control processes typically impose influence at all times t, whereas impulse control acts discretely at times $\tau_1, \tau_2, \dots$. Consequently, an impulse-controlled process often exhibits jump discontinuities.

Definition 1.1. An impulse control $I \equiv \{(\tau_j, \zeta_j), j = 1, \dots, M\}$, where M is a random variable taking values in $\{0, 1, 2, \dots\} \cup \{\infty\}$, consists of:

(a) A sequence of $\{\mathcal{F}_t\}$-stopping times $\tau_1, \tau_2, \dots$ with $0 \le \tau_1 < \tau_2 < \dots$.

(b) A sequence of impulse values $\zeta_1, \zeta_2, \dots$ such that for each j, ζ_j takes values in $Z \subseteq \mathbb{R}^p$ and is measurable with respect to $\mathcal{F}_{\tau_j}$.

The control I is admissible if:

(a) The corresponding state process $X \equiv X^I$ exists and is unique.

(b) With probability one, either $M(\omega) < \infty$ or, if $M(\omega) = \infty$, then $\lim_{j\to\infty} \tau_j(\omega) = \infty$.

We denote by $\mathcal{A}$ the set of all admissible impulse controls.

The condition $\lim_{j\to\infty} \tau_j = \infty$ ensures intervention times do not explode, preventing a reduction to continuous control. The corresponding cost function is defined as:

$$J(x, I) = \mathbb{E}\left[\int_0^\infty e^{-\gamma s} g(X(s))ds + \sum_{j=1}^{\infty} e^{-\gamma\tau_j} \ell(X(\tau_j^-), \zeta_j)\mathbb{1}_{\{\tau_j<\infty\}}\right],$$

where $g : \mathbb{R}^n \mapsto \mathbb{R}$ is the running cost and $\ell : \mathbb{R}^n \times Z \mapsto \mathbb{R}$ is the intervention cost. The value function is $V(x) = \inf_{I\in\mathcal{A}} J(x, I)$.

2 Dynamic Programming for Impulse Control

A key object in impulse control is the intervention operator K. The operator K maps a function $\phi : \mathbb{R}^n \mapsto \mathbb{R}$ to a function $K\phi : \mathbb{R}^n \mapsto \mathbb{R}$ given by:

$$K\phi(x) = \inf_{z\in Z} [\phi(\Gamma(x, z)) + \ell(x, z)], \quad x \in \mathbb{R}^n. \tag{1}$$

Applied to the value function, $KV(x)$ represents the value of the strategy where one takes the best immediate action in state x and behaves optimally thereafter. Clearly, since immediate intervention may not always be optimal, we have $V(x) \le KV(x)$ for all $x \in \mathbb{R}^n$.

To formulate a dynamic programming principle (DPP), let T be an arbitrary finite $\{\mathcal{F}_t\}$-stopping time. If we do not intervene until T, and act optimally afterward, the associated cost is:

$$V(x) \le \mathbb{E}\left[\int_0^T e^{-\gamma s} g(X(s))ds + e^{-\gamma T} KV(X(T^-))\right].$$

If we replace T by T^*, the first time it is optimally required to intervene, the system yields the dynamic programming principle:

$$V(x) = \inf_{T\in\mathcal{S}} \mathbb{E}\left[\int_0^T e^{-\gamma s} g(X(s))ds + e^{-\gamma T} KV(X(T^-))\mathbb{1}_{\{T<\infty\}}\right], \tag{2}$$

where $\mathcal{S}$ is the set of all $\{\mathcal{F}_t\}$-stopping times. Note that Equation (2) heavily resembles the value function of an optimal stopping problem (see Equation (5.1.2) from Ross' notes), with terminal cost h replaced by KV.

Following derivations akin to Section 5.2 from Ross' notes, Ito's rule and limit calculations lead us to the Hamilton-Jacobi-Bellman (HJB) quasi-variational inequality for impulse control:

$$\min\{-\gamma V(x) + \mathcal{L}V(x) + g(x), KV(x) - V(x)\} = 0, \quad x \in \mathbb{R}^n. \tag{3}$$

The states where it is strictly suboptimal to control form the non-intervention region $D = \{x : KV(x) > V(x)\}$. For $x \notin D$, one applies the optimal impulse $z^*(x)$ which minimizes the operator K.

3 Example Impulse Control Problem

Adapted from [Oks03, Example 6.4], consider an optimal dividend payment problem with transaction costs. Without interventions, a cash flow process is a Brownian motion with drift: $X(t) = x + \mu t + \sigma W(t)$. At time $\tau_j \geq 0$, we withdraw $\zeta_j \geq 0$ but pay a cost $c + \lambda\zeta_j$. The state after transaction is $\Gamma(x, z) = x - c - (1 + \lambda)z$. The controlled process is:

$$X(t) = x + \mu t + \sigma W(t) - \sum_{j=1}^{\infty}(c + (1 + \lambda)\zeta_j)\mathbb{1}_{\{\tau_j \leq t\}}.$$

We wish to minimize:

$$J(x, I) = \mathbb{E}\left[\sum_{j=1}^{\infty} e^{-\gamma\tau_j}(-\zeta_j)\mathbb{1}_{\{\tau_j \leq T\}}\right],$$

where $T = \inf\{t \geq 0 : X(t) \leq 0\}$. The value function trivially satisfies $V(x) \leq -(x-c)^+/(1+\lambda)$.

We postulate a non-intervention region $D = \{x : 0 < x < x^*\}$. In D, we seek a solution to:

$$-\gamma\phi(x) + \mu\phi'(x) + \frac{\sigma^2}{2}\phi''(x) = 0.$$

This yields $\phi_0(x) = A(e^{r_+ x} - e^{r_- x})$ where r_+, r_- are the roots of the characteristic equation. For $x \geq x^*$, we find $\hat{z}$ minimizing the intervention operator:

$$K\phi(x) = \inf\left\{\phi(x - c - (1 + \lambda)z) - z : 0 < z \leq \frac{x - c}{1 + \lambda}\right\}.$$

Setting the derivative to zero and applying the principle of smooth fit gives the optimal boundary x^*, the return point $\hat{x}$, and the constant A, fully specifying our candidate value function and optimal control.

4 Connections Between Impulse Control and Optimal Stopping

Recall the DPP from Equation (2). This expression illustrates that we have effectively reduced the impulse control problem to an optimal stopping problem with "terminal cost" KV. While not a pure optimal stopping problem since KV is unknown a priori, this connection facilitates an iterative algorithm.

Let $X^{(0)}$ denote the uncontrolled state process, and let $\phi_0(x) = \mathbb{E}\left[\int_0^\infty e^{-\gamma t} g(X^{(0)}(t))dt\right]$ be the cost with no control. Using the intervention operator K, we recursively define:

$$\phi_k(x) = \inf_{T \in \mathcal{S}} \mathbb{E}\left[\int_0^T e^{-\gamma s} g(X^{(0)}(s))ds + e^{-\gamma T} K\phi_{k-1}(X^{(0)}(T^-))\mathbb{1}_{\{T<\infty\}}\right].$$

Thus, ϕ_k is obtained by solving a series of k optimal stopping problems. Under rigorous mathematical conditions (see [Oks03, Chapter 7]), if $V_k(x)$ denotes the value function restricted to at most k interventions, then $V_k(x) = \phi_k(x)$, and crucially:

$$\lim_{k\to\infty} V_k(x) = V(x), \quad \forall x \in \mathbb{R}^n.$$

By fixing a sufficiently large k, we iteratively construct intervention times τ_j using the stopping regions derived from ϕ_{k-j}, yielding an algorithm that converges to the true optimal impulse control strategy.